Moises Mata

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Planning, control, and flight software for multi-agent planetary robotics

Education

University of Southern California .... Los Angeles, CA
Ph.D. in Aerospace Engineering, advised by Dr. Keenan Albee .... Aug 2026 – Present

Columbia University .... New York, NY
B.A. in Computer Science and Applied Mathematics .... Sept 2022 – May 2026

Honors & Awards

NSF Graduate Research Fellowship Program, Honorable Mention 2026
Adaptive Planning and Control for Heterogeneous Multi-Agent Planetary Exploration

Publications

Mahajan, Ishaan, Khai Nguyen, Sam Schoedel, Elakhya Nedumaran, **Moises Mata**, Brian Plancher, and Zachary Manchester (2026). "Code Generation and Conic Constraints for Model-Predictive Control on Microcontrollers with Conic-TinyMPC". In: *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv: `2403.18149 [cs.RO]`. URL: `https://arxiv.org/abs/2403.18149`.

Research & Work Experience

NASA Jet Propulsion Laboratory, FSW Architecture & Infrastructure .... Pasadena, CA
Intern, Flight Software Engineering .... Jun 2026 – Aug 2026

- Developed long-term "soak-test" infrastructure for the F Prime flight software framework, enabling extended-duration reliability and endurance testing.

Barnard/Dartmouth Accessible and Accelerated Robotics Lab .... New York, NY
Undergraduate Researcher .... Jan 2025 – May 2026

- Diagnosed hardware constraints and modernized the **C++** codebase for TinyMPC, an open-source convex Model Predictive Control solver designed for resource-constrained platforms. Advising professor: Brian Plancher.
- Implemented **meta-learning** algorithms to enable online controller optimization and adaptation across systems with different dynamics.

Columbia SNL Lab .... New York, NY
Undergraduate Researcher .... Sept 2025 – May 2026

- Developed a high-assurance, real-time runtime for spacecraft flight software using **eBPF**, **F Prime**, and **C/C++**, enabling dynamic multithreaded execution of mission logic with strict safety and millisecond deadline guarantees. Advising Professor: Junfeng Yang. (**Publication In Progress**)

NASA Jet Propulsion Laboratory, Small Scale Flight Software Group .... Pasadena, CA
Intern, Flight Software Engineering .... May 2025 – Aug 2025

- Developed core infrastructure for the F Prime flight software framework (v4.0.0 release).
- Architected and implemented reusable software stacks (Communications, Command and Data Handling, Filesystem Support, Data Products), reducing setup time for new missions.

NASA Jet Propulsion Laboratory, Exoplanet Discovery & Science .... Pasadena, CA
Undergraduate Research Fellow .... May 2024 – Aug 2024

- Trained binary classification models using **Python**, **scikit-learn**, within **Jupyter Notebook** to predict the presence of habitable planets from simulated observational data.

Leadership Experience

Columbia Space Initiative .... New York, NY
Executive Board: Co-President (2025–26), Treasurer (2024–25) .... March 2024 – May 2026

- Led **Columbia's largest engineering club** (**250+ active members**, **$200k annual budget**) as **Co-President**, coordinating strategy across **13 active projects**.
- Organized major events including company visits, faculty talks, and astronaut visits, engaging Columbia and the broader NYC community in aerospace.
- Conducted career development workshops (resume/CV, internships) for Columbia's undergraduate engineering population and led external outreach to **1000+ middle and high school students**, primarily translating engineering concepts into Spanish to inspire underrepresented demographics.

Projects

Flight Software Lead, LIONESS (6U CubeSat) .... Aug 2024 – May 2026

- Scientific mission to image the circumgalactic medium of nearby galaxies in H-alpha spectrum. Spectrograph instrument developed with Schiminovich Astronomy Instrumentation Lab, Advised by Professor David Schiminovich.
- Architected flight software infrastructure for a science-driven mission. Co-authored NASA Astrophysics Research and Analysis proposal (rated **"Selectable"**); manifested via **NASA CubeSat Launch Initiative**.

Flight Software Lead, PROVES Alcyone (1U CubeSat) .... Aug 2024 – May 2026

- **Columbia's first satellite**, launched April 2026 to capture Earth imagery. Implemented **F Prime** components for watchdog, power management, and offboard camera control via UART.

NASA SUITS Mission Co-Lead .... Sept 2023 – Jul 2024

- Co-authored proposal selected as one of **17 national finalists**; developed an AR astronaut-assistance interface in **Unity (C#)** for **Microsoft HoloLens 2** and led testing at **NASA Johnson Space Center**.

Skills

Languages	C, C++, C#, Python, MATLAB, Julia
Frameworks & Tools	F Prime, Unity, scikit-learn, CircuitPython, Jupyter
Focus Areas	Robotic Optimization & Control, Embedded Systems, Flight Software
Other	English (native), Spanish (native), French (limited); Amateur Radio (Technician)